

HW 0513

$$\frac{300 \times 447}{900} = 166$$

$$\frac{300 \times 402}{900} = 134$$

10.

Level	Married	No. married	
Bachelor or higher	187 ($\frac{100 \times 447}{900}$)	113 ($\frac{100 \times 402}{900}$)	300
College	162 ($\frac{100 \times 447}{900}$)	138 ($\frac{100 \times 402}{900}$)	300
High school or less	144 ($\frac{100 \times 447}{900}$)	151 ($\frac{100 \times 402}{900}$)	300
	493	402	900

H_0 : education level & married is independent

H_a : / / (1) " dependent

$$\chi^2 = \frac{(187-166)^2}{166} + \frac{(162-166)^2}{166} + \frac{(144-166)^2}{166} + \frac{(113-134)^2}{134} + \frac{(138-134)^2}{134} + \frac{(151-134)^2}{134}$$

$$= 12.0611$$

$\chi^2_{2,0.05} = 5.991 < \chi^2$, there is sufficient evidence to reject H_0 .

11.

	Type within			
Family number	Apartment	duplex	single within	
1	8 (9.67)	2 (9.67)	1 (9.67)	21
2	16 (11)	8 (11)	9 (11)	33
3	10 (11.33)	10 (11.33)	14 (11.33)	34
4/total	6 (8)	2 (8)	16 (8)	24
	40	40	40	120

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i} = \frac{(8-9.67)^2}{9.67} + \frac{(2-9.67)^2}{9.67} + \frac{(1-9.67)^2}{9.67} + \frac{(16-11)^2}{11} + \frac{(8-11)^2}{11} + \frac{(9-11)^2}{11} + \frac{(10-11.33)^2}{11.33} + \frac{(10-11.33)^2}{11.33} + \frac{(14-11.33)^2}{11.33} + \frac{(6-8)^2}{8} + \frac{(2-8)^2}{8} + \frac{(16-8)^2}{8}$$

$$= 36.449$$

$\chi^2_{6,0.01} = 16.812 < \chi^2$, reject H_0

$\therefore$ 4 types of apartment, duplex, single, high school or less, high school or less, high school or less

①

12) ~~if~~

A	4.7	5.2	5	4.2	5.1
B	5	4.7	4.9	4.1	4.9

$$\bar{x}_A = 5, s_A^2 = \frac{\sum (x_i - \bar{x})^2}{n_1 - 1} = 0.025$$

$$\bar{x}_B = 4.8, s_B^2 = 0.013$$

a) ~~sum~~ rank sum test

	B	B	A	A	B	B	A	A	A
	4.7	4.8	4.9	4.9	4.9	5	5	5.1	5.2
rank	1	2.5	2.5	4.5	4.5	5	7.5	7.5	10

$$T_A = 2.5 + 5 + 7.5 + 9 + 10 = 34$$

H_0 : 上风口与下风口

H_1 : 上风口空气量 > 下风口

$$T_A^* = 5(11) - 34 = 21$$

$$b) H_0: \mu_1 - \mu_2 = 0$$

$$H_1: \mu_1 - \mu_2 > 0$$

$$\min(T_A, T_A^*) = T_A^* = 21$$

$$t = \frac{\bar{x}_A - \bar{x}_B}{\sqrt{s_p^2 (\frac{1}{n_1} + \frac{1}{n_2})}} = \frac{5 - 4.8}{\sqrt{0.019 (\frac{1}{5} + \frac{1}{5})}} = 1.606$$

$$t_{0.05, 10} = 1.812$$

so we reject H_0

$$T_{5, 10, 0.05} = 19$$

$\therefore$ reject H_0

recall p-value

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2} = \frac{4(0.025) + 4(0.013)}{8} = 0.019$$

13.

	99	99
	D	H
	2.75(15)	0.29(1)
	2.14(11)	6.43(7)
	3.23(8)	1.06(4)
	2.17(10)	1.01(1)
	2.44(14)	0.04(2)
	2.19(12)	1.47(6)
	3.16(17)	1.12(5)
	2.11(16)	2.01(9)
	2.2(13)	1.12(5.5)
Rank sum	126	48

so we rank sum test

$$T_D = 126$$

$$T_D^* = 9(9+9+1) - 126 = 9(19) - 126 = 45$$

$$T_{9, 18, 0.05} = 62 \text{ reject } H_0$$

sign test

$$\alpha = 0.1$$

	p_{ij}						
i	1	2	3	4	5	6	7
1	8.84	1.1	9.1	7.7	10.4	5.1	7.4
2	8.8	7.4	9.0	7.8	9.9	1.1	6.9

sign
1-2
(+) (+) (+) (-) (+) (+) (+)

$$X \sim \text{bin}(7, 0.5)$$

$$T^- = 1$$

H_0 : same distribution

H_1 : not same distribution

$$\text{so } P(X \leq 1) = P(X=0) + P(X=1)$$

$$= C_0^7 (0.5)^7 + C_1^7 (0.5)^7$$

$$= 0.00781 + 7(0.00781)$$

$$= 0.06247$$

$$p\text{-value} = 2 P(X \leq 1)$$

$$= 2 (0.06247)$$

$$= 0.12494 > \alpha$$

$\therefore$ we accept H_0

RR. X	p
0	0.0078
1	0.06247
2	0.22781641
3	0.22781
4	0.06247
5	0.0078
6	0.0078
7	0.0078

$$\frac{1}{2} \{0, 7\} = 2(0.0078)$$

$$= 0.0156 < \alpha = 0.1$$

$$\{0, 1, 6, 7\} = 0.0156 + 2(0.06247)$$

$$= (0.12494 + 0.0156) \times 2$$

$$= 0.12494 > \alpha = 0.1$$

$$\therefore \text{RR } Z = \{0, 7\}$$

$$\text{we reject } X \geq 1$$

11. Number of CT) = 11

Number of C) = 1

$\alpha = 0.05$

So we have



5

a) $H_0: p = 0.5$ $H_1: p \neq 0.5$

$X \sim \text{Bin}(20, 0.5)$

$P(X \leq 5) = 0.0207$ $P(X \leq 6) = 0.0577$

$\alpha = 0.05$

So Rejection region = $\{X \leq 5 \text{ or } X \geq 15\}$

$X = 11$, not in RE, not reject

b) $Z = \frac{\hat{p} - p}{\sqrt{\frac{p(1-p)}{n}}} = \frac{11 - 20(0.5)}{\sqrt{\frac{0.5 \cdot 0.5}{20}}}$

$= 0.447 < Z_{0.025}$

$= 1.96$

c) a & b for same test

∴ not reject

11. $\frac{2}{3}$ percentage recovered in 62

a) = 2

b) = 8

$n = 10$

$H_0: p = 0.5$

$H_1: p \neq 0.5$

$X \sim \text{Bin}(10, 0.5)$

$\alpha = 0.05$

So $P(X \leq 2) + P(X \geq 8) = 0.022$

$P(X \leq 1) + P(X \geq 9)$

$P(X \geq 0) = C_{10}^0 (0.5)^{10} = 0.001$

$P(X = 1) = C_{10}^1 (0.5)^{10} = 0.0098$

$P(X = 2) = C_{10}^2 (0.5)^{10} = 0.0474$

$2(P \leq 1) = 2(0.001 + 0.0098) = 0.0216$

$2(P \leq 2) = 2(0.001 + 0.0098 + 0.0474)$

$= 0.1094 > \alpha = 0.05$

So RE: $P(X \leq 1) + P(X \geq 9)$
 $X \leq 1 \text{ \& } X \geq 9$

$X = 2$

So we reject.

b) 2 is not in rejection region, not reject
sign test can be used
to reject.

4)